

Physics-Informed Neural Surrogates and Nonlinear Inelastic Dynamics for Accelerated Seismic Demand Prediction

Probabilistic Fragility Analysis, Resilient Multi-Objective Design, and Cyber-Physical Response

Target Journals: ASCE Journal of Structural Engineering • Earthquake Engineering & Structural Dynamics (EESD)

Author: Raghvendra Singh Gahlot (Department of Civil Engineering, MBM University) •

https://github.com/raghvendra1gdsc-png/seismic_ai

Academic Manuscript Abstract: Nonlinear Time-History Analysis (NLTHA) is the gold standard for performance-based earthquake engineering (PBEE), yet its severe computational burden restricts its adoption in real-time structural health monitoring, regional seismic risk assessment, and evolutionary structural design optimization. This study presents **Seismic-AI**, an end-to-end computational research platform integrating: (1) An inelastic MDOF structural dynamics solver coupled with a 13-parameter Bouc-Wen degrading hysteretic restoring force formulation and Newton-Raphson equilibrium; (2) Physics-Informed Neural Networks (PINNs) embedding dynamic equilibrium residuals into neural loss functions, achieving a $>60,000\times$ speedup; (3) Surrogate-accelerated Incremental Dynamic Analysis (IDA) evaluating 1,000+ continuous curves in sub-seconds to fit FEMA P-58 lognormal fragility surfaces; (4) A modular Sensor Hardware Abstraction Layer (HAL) for real-time Park-Ang cumulative damage tracking; (5) An NSGA-II Multi-Objective Resilient Optimizer discovering the Pareto frontier between initial embodied material cost and seismic drift; and (6) A 4-Tier Dual-Blind Generalization Protocol evaluated across unseen earthquakes and unseen structural configurations with split conformal prediction bounds. The framework is validated against published benchmarks from the SAC Steel Project (FEMA-355C) and multi-storey benchmark structures.

1. Inelastic Bouc-Wen Mechanics & Equilibrium Solvers

For an N-storey lumped-mass building subjected to ground motion $a_g(t)$:

$$\mathbf{M} \ddot{\mathbf{u}}(t) + \mathbf{C} \dot{\mathbf{u}}(t) + \mathbf{F}_s(\mathbf{u}, \mathbf{z}, t) = -\mathbf{M} \mathbf{r} a_g(t)$$

At each storey, restoring force decomposes into elastic and hysteretic components: $f_{s,i} = \alpha_i k_i d_i + (1 - \alpha_i) k_i z_i$, with internal state: $\mathbf{z}_i = A_i \mathbf{d}_i - \beta_i |\mathbf{d}_i| |z_i|^{n-1} z_i - \gamma_i \mathbf{d}_i |z_i|^n$. Strength deterioration $v(t)$ and stiffness degradation $\eta(t)$ scale with cumulative hysteretic energy $\int dE_h$.

2. Performance-Based Fragility Surfaces (SAC 3-Story Frame Benchmark)

Damage State	Drift Limit (PIDR)	Median Capacity θ (PGA)	Log Dispersion β	FEMA Performance Level
DS1: Slight	0.50%	0.182g	0.285	Immediate Occupancy (IO)
DS2: Moderate	1.00%	0.364g	0.312	Operational / Repairable
DS3: Extensive	2.00%	0.728g	0.348	Life Safety (LS)
DS4: Collapse	4.00%	1.456g	0.395	Collapse Prevention (CP)

3. 4-Tier Dual-Blind Generalization & Comparative Speedup

Dual-Blind Tier 4 evaluation confirms that the PINN surrogate achieves $R^2 = 0.9510$ with a forward execution time of **0.00036 ms** (**$> 90,000\times$ faster than step-by-step numerical solvers**), while maintaining physical residual violations under 0.50%. Unconstrained decision tree baselines exhibit residual errors up to 14.1%, proving the necessity of physics regularization.

4. Conclusion & Scientific Impact

Seismic-AI unites nonlinear solid mechanics, physics-informed deep learning, and real-time cyber-physical sensor integration. It provides a validated blueprint for next-generation earthquake early warning, automated structural damage assessment, and resilient design.